

```
In [26]: # Copyright 2026 Google LLC
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```

Multimodal Retrieval Augmented Generation (RAG) using Gemini API in Agent Platform



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Overview

Retrieval augmented generation (RAG) has become a popular paradigm for enabling LLMs to access external data and also as a mechanism for grounding to mitigate against hallucinations.

In this notebook, you will learn how to perform multimodal RAG where you will perform Q&A over a financial document filled with both text and images.

Gemini

Gemini is a family of generative AI models developed by Google DeepMind that is designed for multimodal use cases. The Gemini API gives you access to the Gemini models.

Comparing text-based and multimodal RAG

Multimodal RAG offers several advantages over text-based RAG:

1. **Enhanced knowledge access:** Multimodal RAG can access and process both textual and visual information, providing a richer and more comprehensive knowledge base for the LLM.
2. **Improved reasoning capabilities:** By incorporating visual cues, multimodal RAG can make better informed inferences across different types of data modalities.

This notebook shows you how to use RAG with Gemini API in Agent Platform, [text embeddings](#), and [multimodal embeddings](#), to build a document search engine.

Through hands-on examples, you will discover how to construct a multimedia-rich metadata repository of your document sources, enabling search, comparison, and reasoning across diverse information streams.

Objectives

This notebook provides a guide to building a document search engine using multimodal retrieval augmented generation (RAG), step by step:

1. Extract and store metadata of documents containing both text and images, and generate embeddings the documents
2. Search the metadata with text queries to find similar text or images
3. Search the metadata with image queries to find similar images
4. Using a text query as input, search for contextual answers using both text and images

Costs

This tutorial uses billable components of Google Cloud:

- Agent Platform

Learn about [Agent Platform pricing](#) and use the [Pricing Calculator](#) to generate a cost estimate based on your projected usage.

Getting Started

Install Gen AI SDK for Python and other dependencies

```
In [27]: %pip install --upgrade --quiet google-genai
          %pip install --quiet pymupdf
```

```

WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
Note: you may need to restart the kernel to use updated packages.
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution ~oogle-auth (/opt/micromamba/lib/python3.12/site-packages)
Note: you may need to restart the kernel to use updated packages.

```

Restart current runtime

To use the newly installed packages in this Jupyter runtime, you must restart the runtime. You can do this by running the cell below, which will restart the current kernel.

```

In [28]: # Restart kernel after installs so that your environment can access the n
import IPython

app = IPython.Application.instance()
app.kernel.do_shutdown(True)

```

```

Out[28]: {'status': 'ok', 'restart': True}

```

Define Google Cloud project information

```

In [2]: from google import genai
import utils.intro_multimodal_rag_utils as utils

PROJECT_ID = "qwiklabs-gcp-02-288e6f2d8eed" # @param {type:"string"}
LOCATION = "global"

client = genai.Client(enterprise=True, project=PROJECT_ID, location=LOCAT
utils.set_global_client(client)

```

Import libraries

```

In [3]: from IPython.display import Markdown, display
from rich.markdown import Markdown as rich_Markdown
from google.genai import types
from PIL import Image as PIL_Image

```

Load the Gemini model

```
In [4]: model = "gemini-3.1-flash-lite"
```

Download custom Python utilities & required files

The cell below will download a helper functions needed for this notebook, to improve readability. It also downloads other required files. You can also view the code for the utils here: (`intro_multimodal_rag_utils.py`) directly on [GitHub](#).

```
In [5]: # download documents and images used in this notebook
!gsutil -m rsync -x ".*_utils\.py$" -r gs://github-repo/rag/intro_multimo
print("Download completed")
```

Google recommends using Gcloud storage CLI (<https://docs.cloud.google.com/storage/docs/discover-object-storage-gcloud>) instead of gsutil. Please refer to migration guide (<https://docs.cloud.google.com/storage/docs/gsutil-transition-to-gcloud>) for assistance.

Building synchronization state...

Starting synchronization...

Download completed

Building metadata of documents containing text and images

The data

The source data that you will use in this notebook is a modified version of [Google-10K](#) which provides a comprehensive overview of the company's financial performance, business operations, management, and risk factors. As the original document is rather large, you will be using a modified version with only 14 pages, split into two parts - [Part 1](#) and [Part 2](#) instead. Although it's truncated, the sample document still contains text along with images such as tables, charts, and graphs.

Import helper functions to build metadata

Before building the multimodal RAG system, it's important to have metadata of all the text and images in the document. For references and citations purposes, the metadata should contain essential elements, including page number, file name, image counter, and so on. Hence, as a next step, you will generate embeddings from the metadata, which will be required to perform similarity search when querying the data.

```
In [6]: from utils.intro_multimodal_rag_utils import get_document_metadata
```

Extract and store metadata of text and images from a document

You just imported a function called `get_document_metadata()`. This function extracts text and image metadata from a document, and returns two dataframes, namely `text_metadata` and `image_metadata`, as outputs. If you want to find out more about how `get_document_metadata()` function is implemented using Gemini and the embedding models, you can take look at the [source code](#) directly.

The reason for extraction and storing both text metadata and image metadata is that just by using either of the two alone is not sufficient to come out with a relevant answer. For example, the relevant answers could be in visual form within a document, but text-based RAG won't be able to take into consideration of the visual images. You will also be exploring this example later in this notebook.

At the next step, you will use the function to extract and store metadata of text and images from a document. Please note that the following cell may take a few minutes to complete:

Note:

The current implementation works best:

- if your documents are a combination of text and images.
- if the tables in your documents are available as images.
- if the images in the document don't require too much context.

Additionally,

- If you want to run this on text-only documents, use normal RAG
- If your documents contain particular domain knowledge, pass that information in the prompt below.

⚠ Do not send more than 50 pages in the logic below, its not designed to do that and you will get into quota issue. ⚠

```
In [7]: # Specify the PDF folder with multiple PDF

# pdf_folder_path = "/content/data/" # if running in Google Colab/Colab E
pdf_folder_path = "data/" # if running in Agent Platform Workbench.

# Specify the image description prompt. Change it
image_description_prompt = """Explain what is going on in the image.
If it's a table, extract all elements of the table.
If it's a graph, explain the findings in the graph.
Do not include any numbers that are not mentioned in the image.
"""

# Extract text and image metadata from the PDF document
text_metadata_df, image_metadata_df = get_document_metadata(
    model, # we are passing "gemini-3.1-flash-lite" model
    pdf_folder_path,
    image_save_dir="images",
    image_description_prompt=image_description_prompt,
```

```

    embedding_size=1408,
    # add_sleep_after_page = True, # Uncomment this if you are running in
    # sleep_time_after_page = 5,
    # generation_config = # see next cell
    # safety_settings = # see next cell
)

print("\n\n --- Completed processing. ---")

```

Processing the file: ----- data/google-10k-sample-part2.pdf

```

Processing page: 1
Extracting image from page: 1, saved as: images/google-10k-sample-part2.pdf_image_0_0_6.jpeg
Extracting image from page: 1, saved as: images/google-10k-sample-part2.pdf_image_0_1_8.jpeg
Processing page: 2
Extracting image from page: 2, saved as: images/google-10k-sample-part2.pdf_image_1_0_13.jpeg
Processing page: 3
Processing page: 4
Extracting image from page: 4, saved as: images/google-10k-sample-part2.pdf_image_3_0_19.jpeg
Processing page: 5
Extracting image from page: 5, saved as: images/google-10k-sample-part2.pdf_image_4_0_22.jpeg
Extracting image from page: 5, saved as: images/google-10k-sample-part2.pdf_image_4_1_23.jpeg
Processing page: 6
Extracting image from page: 6, saved as: images/google-10k-sample-part2.pdf_image_5_0_26.jpeg
Processing page: 7

```

Processing the file: ----- data/google-10k-sample-part1.pdf

```

Processing page: 1
Processing page: 2
Extracting image from page: 2, saved as: images/google-10k-sample-part1.pdf_image_1_0_11.jpeg
Processing page: 3
Extracting image from page: 3, saved as: images/google-10k-sample-part1.pdf_image_2_0_15.jpeg
Processing page: 4
Extracting image from page: 4, saved as: images/google-10k-sample-part1.pdf_image_3_0_18.jpeg
Processing page: 5
Extracting image from page: 5, saved as: images/google-10k-sample-part1.pdf_image_4_0_21.jpeg
Processing page: 6
Processing page: 7

```

--- Completed processing. ---

```
In [8]: ## Parameters for Gemini API call.
## reference for parameters: https://google.github.io/generative-ai-

# generation_config = types.GenerateContentConfig(temperature=0.2, max_ou

## Set the safety settings if Gemini is blocking your content or you are
## ref for settings and thresholds: https://google.github.io/generative-

# safety_settings = [
#     types.SafetySetting(
#         category=types.HarmCategory.HARM_CATEGORY_HARASSMENT,
#         threshold=types.HarmBlockThreshold.BLOCK_NONE,
#     ),
#     types.SafetySetting(
#         category=types.HarmCategory.HARM_CATEGORY_HATE_SPEECH,
#         threshold=types.HarmBlockThreshold.BLOCK_NONE,
#     ),
#     types.SafetySetting(
#         category=types.HarmCategory.HARM_CATEGORY_SEXUALLY_EXPLICIT,
#         threshold=types.HarmBlockThreshold.BLOCK_NONE,
#     ),
#     types.SafetySetting(
#         category=types.HarmCategory.HARM_CATEGORY_DANGEROUS_CONTENT,
#         threshold=types.HarmBlockThreshold.BLOCK_NONE,
#     ),
# ]

## You can also pass parameters and safety_setting to "get_gemini_respon
```

Inspect the processed text metadata

The following cell will produce a metadata table which describes the different parts of text metadata, including:

- **text**: the original text from the page
- **text_embedding_page**: the embedding of the original text from the page
- **chunk_text**: the original text divided into smaller chunks
- **chunk_number**: the index of each text chunk
- **text_embedding_chunk**: the embedding of each text chunk

```
In [9]: text_metadata_df.head()
```

Out [9]:

	file_name	page_num	text	text_embedding_
0	google-10k-sample-part2.pdf	1	source: https://abc.xyz/assets/investor/static...	[0.01718468777835 -0.009448662400245
1	google-10k-sample-part2.pdf	2	APAC revenue growth from 2020 to 2021 was favo...	[-0.005425038747489 -0.0269870441406965
2	google-10k-sample-part2.pdf	2	APAC revenue growth from 2020 to 2021 was favo...	[-0.005425038747489 -0.0269870441406965
3	google-10k-sample-part2.pdf	3	increases in content acquisition costs primari...	[-0.02321242354810 -0.02880910225212574
4	google-10k-sample-part2.pdf	3	increases in content acquisition costs primari...	[-0.02321242354810 -0.02880910225212574

Inspect the processed image metadata

The following cell will produce a metadata table which describes the different parts of image metadata, including:

- **img_desc**: Gemini-generated textual description of the image.
- **mm_embedding_from_text_desc_and_img**: Combined embedding of image and its description, capturing both visual and textual information.
- **mm_embedding_from_img_only**: Image embedding without description, for comparison with description-based analysis.
- **text_embedding_from_image_description**: Separate text embedding of the generated description, enabling textual analysis and comparison.

In [10]:

image_metadata_df.head()

Out[10]:

	file_name	page_num	img_num	img_path	img_desc	mm_emb
0	google-10k-sample-part2.pdf	1	1	images/google-10k-sample-part2.pdf_image_0_0_6...	The image displays a table showing the percent...	[-0.0
1	google-10k-sample-part2.pdf	1	2	images/google-10k-sample-part2.pdf_image_0_1_8...	The image displays a financial table comparing...	[0.0
2	google-10k-sample-part2.pdf	2	1	images/google-10k-sample-part2.pdf_image_1_0_1...	The image displays a financial table comparing...	[-0.00
3	google-10k-sample-part2.pdf	4	1	images/google-10k-sample-part2.pdf_image_3_0_1...	The image displays a financial table detailing...	[0.0
4	google-10k-sample-part2.pdf	5	1	images/google-10k-sample-part2.pdf_image_4_0_2...	The image displays a financial table detailing...	[0.

Import the helper functions to implement RAG

You will be importing the following functions which will be used in the remainder of this notebook to implement RAG:

- **get_similar_text_from_query():** Given a text query, finds text from the document which are relevant, using cosine similarity algorithm. It uses text embeddings from the metadata to compute and the results can be filtered by top score, page/chunk number, or embedding size.
- **print_text_to_text_citation():** Prints the source (citation) and details of the retrieved text from the `get_similar_text_from_query()` function.
- **get_similar_image_from_query():** Given an image path or an image, finds images from the document which are relevant. It uses image embeddings from the metadata.
- **print_text_to_image_citation():** Prints the source (citation) and the details of retrieved images from the `get_similar_image_from_query()` function.
- **get_gemini_response():** Interacts with a Gemini model to answer questions based on a combination of text and image inputs.
- **display_images():** Displays a series of images provided as paths or PIL Image objects.

In [11]: `from utils.intro_multimodal_rag_utils import (
display_images,`

```
get_gemini_response,  
get_similar_image_from_query,  
get_similar_text_from_query,  
print_text_to_image_citation,  
print_text_to_text_citation,  
)
```

Before implementing a multimodal RAG, let's take a step back and explore what you can achieve with just text or image embeddings alone. It will help to set the foundation for implementing a multimodal RAG, which you will be doing in the later part of the notebook. You can also use these essential elements together to build applications for multimodal use cases for extracting meaningful information from the document.

Text Search

Let's start the search with a simple question and see if the simple text search using text embeddings can answer it. The expected answer is to show the value of basic and diluted net income per share of Google for different share types.

```
In [12]: query = "I need details for basic and diluted net income per share of Cla
```

Search similar text with text query

```
In [13]: # Matching user text query with "chunk_embedding" to find relevant chunks  
matching_results_text = get_similar_text_from_query(  
    query,  
    text_metadata_df,  
    column_name="text_embedding_chunk",  
    top_n=3,  
    chunk_text=True,  
)  
  
# Print the matched text citations  
print_text_to_text_citation(matching_results_text, print_top=False, chunk
```

Citation 1: Matched text:**Score:** 0.81**File name:** google-10k-sample-part2.pdf**Page number:** 4**Chunk number:** 1**Chunk text:** liquidation and dividend rights are identical, the undistributed earnings are

allocated on a proportionate basis.

In the years ended December 31, 2019, 2020 and 2021, the net income per share amounts are the same for Class A, Class B, and Class C stock because the holders of each class are entitled to equal per share dividends or distributions

in liquidation in accordance with the Amended and Restated Certificate of Incorporation of Alphabet Inc.

The following tables set forth the computation of basic and diluted net income per

share of Class A, Class B, and Class C stock (in millions, except share amounts

which are reflected in thousands and per share amounts):

Citation 2: Matched text:**Score:** 0.77**File name:** google-10k-sample-part2.pdf**Page number:** 3**Chunk number:** 1**Chunk text:** increases in content acquisition costs primarily for YouTube, data center and

other operations costs, and hardware costs. The increase in data center and

Table of Contents Alphabet Inc. 36 other operations costs was partially offset by

a reduction in depreciation expense due to the change in the estimated useful life

of our servers and certain network equipment beginning in the first quarter of

2021.

Net Income Per Share

We compute net income per share of Class A, Class B, and Class C stock using

the two-class method. Basic net income per share is computed using the

weighted-average number of shares outstanding during the period. Diluted net

income per share is computed using the weighted-average number of shares and

the effect of potentially dilutive securities outstanding during the period.

Potentially dilutive securities consist of restricted stock units and other

contingently issuable shares. The dilutive effect of outstanding restricted stock

units and other contingently issuable

Citation 3: Matched text:**Score:** 0.71**File name:** google-10k-sample-part1.pdf**Page number:** 1**Chunk number:** 1**Chunk text:** source: https://abc.xyz/assets/investor/static/pdf/20220202_alphabet_10K.pdf

source: <https://abc.xyz/assets/9a/bd/838c917c4b4ab21f94e84c3c2c65/goog-10-k-q4-2022.pdf>

Note: Tables and figures are converted to images for demonstration purpose S.

MARKET FOR REGISTRANTS COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

As of October 2, 2015, Alphabet Inc. became the successor issuer of Google Inc.

pursuant to Rule 12g-3(a) under the Exchange Act. Our Class A common stock has been listed on the Nasdaq Global Select Market under the symbol GOOG since August 19, 2004 and under the symbol "GOOGL" since April 3, 2014. Prior

to August 19, 2004, there was no public market for our stock. Our Class B common stock is neither listed nor traded. Our Class C capital stock has been

listed on the Nasdaq Global Select Market under the symbol GOOG since April

3, 2014.

Holder of Record

As of December 31, 2021, there were approximately 4,907 and 1,733 stockholders of record

You can see that the first high score match does have what we are looking for, but upon closer inspection, it mentions that the information is available in the "following" table. The table data is available as an image rather than as text, and hence, the chances are you will miss the information unless you can find a way to process images and their data.

However, Let's feed the relevant text chunk across the data into the Gemini model and see if it can get your desired answer by considering all the chunks across the document. This is like basic text-based RAG implementation.

```
In [14]: print("\n **** Result: ***** \n")

# All relevant text chunk found across documents based on user query
context = "\n".join(
    [value["chunk_text"] for key, value in matching_results_text.items()]
)

instruction = f"""Answer the question with the given context.
If the information is not available in the context, just return "not available"
Question: {query}
Context: {context}
Answer:
"""

# Prepare the model input
model_input = instruction

# Generate Gemini response with streaming output
get_gemini_response(
    model, # we are passing Gemini
    model_input=model_input,
    stream=True,
    generation_config=types.GenerateContentConfig(temperature=0.2),
)

**** Result: *****
```

```
Out[14]: 'not available in the context'
```

You can see that it returned:

```
"not available in the context"
```

This is expected as discussed previously. No other text chunk (total 3) had the information you sought. This is because the information is only available in the images rather than in the text part of the document. Next, let's see if you can solve this problem by leveraging Gemini and Multimodal Embeddings.

Note: We handcrafted examples in our document to simulate real-world cases where information is often embedded in charts, table, graphs, and other image-based elements and unavailable as plain text.

Search similar images with text query

Since plain text search didn't provide the desired answer and the information may be visually represented in a table or another image format, you will use multimodal capability of Gemini model for the similar task. The goal here also is to find an image similar to the text query. You may also print the citations to verify.

```
In [15]: query = "I need details for basic and diluted net income per share of Cla
```

```
In [16]: matching_results_image = get_similar_image_from_query(
    text_metadata_df,
    image_metadata_df,
    query=query,
    column_name="text_embedding_from_image_description", # Use image des
    image_emb=False, # Use text embedding instead of image embedding
    top_n=3,
    embedding_size=1408,
)

# Markdown(print_text_to_image_citation(matching_results_image, print_top
print("\n **** Result: ***** \n")

# Display the top matching image
display(matching_results_image[0]["image_object"])

**** Result: *****
```

	Year Ended December 31,		
	2020		
	Class A	Class B	Class C
Basic net income per share:			
Numerator			
Allocation of undistributed earnings	\$ 17,733	\$ 2,732	\$ 19,804
Denominator			
Number of shares used in per share computation	299,815	46,182	334,819
Basic net income per share	<u>\$ 59.15</u>	<u>\$ 59.15</u>	<u>\$ 59.15</u>
Diluted net income per share:			
Numerator			
Allocation of undistributed earnings for basic computation	\$ 17,733	\$ 2,732	\$ 19,804
Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares	2,732	0	0
Reallocation of undistributed earnings	(180)	(25)	180
Allocation of undistributed earnings	<u>\$ 20,285</u>	<u>\$ 2,707</u>	<u>\$ 19,984</u>
Denominator			
Number of shares used in basic computation	299,815	46,182	334,819
Weighted-average effect of dilutive securities			
Add:			
Conversion of Class B to Class A shares outstanding	46,182	0	0
Restricted stock units and other contingently issuable shares	87	0	6,125
Number of shares used in per share computation	<u>346,084</u>	<u>46,182</u>	<u>340,944</u>
Diluted net income per share	<u>\$ 58.61</u>	<u>\$ 58.61</u>	<u>\$ 58.61</u>

Bingo! It found exactly what you were looking for. You wanted the details on Google's Class A, B, and C shares' basic and diluted net income, and guess what? This image fits the bill perfectly thanks to its descriptive metadata using Gemini.

You can also send the image and its description to Gemini and get the answer as JSON:

```
In [17]: print("\n **** Result: ***** \n")

# All relevant text chunk found across documents based on user query
context = f"""Image: {matching_results_image[0]['image_object']}
Description: {matching_results_image[0]['image_description']}
****

instruction = f"""Answer the question in JSON format with the given conte
Question: {query}
Context: {context}
Answer:
****

# Prepare the model input
model_input = instruction

# Generate Gemini response with streaming output
Markdown(
    get_gemini_response(
        model, # we are passing "gemini-3.1-flash-lite" model
        model_input=model_input,
        stream=True,
        generation_config=types.GenerateContentConfig(temperature=1),
    )
)

**** Result: *****
```

```
Out [17]: { "Basic net income per share": { "Class A": "$ 59.15", "Class B": "$ 59.15", "Class C":  
"$ 59.15" }, "Diluted net income per share": { "Class A": "$ 58.61", "Class B": "$  
58.61", "Class C": "$ 58.61" } }
```

```
In [18]: ## you can check the citations to probe further.  
## check the "image description:" which is a description extracted through  
Markdown(print_text_to_image_citation(matching_results_image, print_top=10))
```

Citation 1:

Matched image path, page number, and page text:

Score: 0.76

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_4_0_22.jpeg

Page number: 5

Page text: Stock-Based Award Activities

The weighted-average grant-date fair value of RSUs granted during the year s

ended December 31, 2019 and 2020 was \$1,092.36 and \$1,407.97, respectively.

Total fair value of RSUs, as of their respective vesting dates, during the years

ended December 31, 2019, 2020, and 2021 were \$15.2 billion, \$17.8 billion, and

\$28.8 billion, respectively. As of December 31, 2021, there was \$25.8 billion of

unrecognized compensation cost related to unvested employee RSUs. This amount is expected to be recognized over a weighted-average period of 2.5 years. 401(k) Plans We have two 401(k) Savings Plans that qualify as deferred

Image description: The image displays a financial table detailing the calculation of basic and diluted net income per share for three different classes of stock (Class A, Class B, and Class C) for the year ended December 31, 2020.

Table Data

Basic net income per share:

* **Numerator:**

* Allocation of undistributed earnings: Class A (\$ 17,733), Class B (\$ 2,732), Class C (\$ 19,804)

* **Denominator:**

* Number of shares used in per share computation: Class A (299,815), Class B (46,182), Class C (334,819)

* **Basic net income per share:** Class A (\$ 59.15), Class B (\$ 59.15), Class C (\$ 59.15)

Diluted net income per share:

* **Numerator:**

* Allocation of undistributed earnings for basic computation: Class A (\$ 17,733), Class B (\$ 2,732), Class C (\$ 19,804)

* Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares: Class A (2,732), Class B (0), Class C (0)

* Reallocation of undistributed earnings: Class A ((180)), Class B ((25)), Class C (180)

* Allocation of undistributed earnings: Class A (\$ 20,285), Class B (\$ 2,707), Class C (\$ 19,984)

* **Denominator:**

* Number of shares used in basic computation: Class A (299,815), Class B (46,182), Class C (334,819)

* **Weighted-average effect of dilutive securities:**

* Add: Conversion of Class B to Class A shares outstanding: Class A (46,182), Class B (0), Class C (0)

* Add: Restricted stock units and other contingently issuable shares: Class A (87), Class B (0), Class C (6,125)

* Number of shares used in per share computation: Class A (346,084), Class B (46,182), Class C (340,944)

```
* **Diluted net income per share:** Class A ($ 58.61), Class B ($ 58.61), Class C ($ 58.61)
```

```
Out[18]: <IPython.core.display.Markdown object>
```

Image Search

Search similar image with image query

Imagine searching for images, but instead of typing words, you use an actual image as the clue. You have a table with numbers about the cost of revenue for two years, and you want to find other images that look like it, from the same document or across multiple documents.

Think of it like searching with a mini-map instead of a written address. It's a different way to ask, "Show me more stuff like this". So, instead of typing "cost of revenue 2020 2021 table", you show a picture of that table and say, "Find me more like this"

For demonstration purposes, we will only be finding similar images that show the cost of revenue or similar values in a single document below. However, you can scale this design pattern to match (find relevant images) across multiple documents.

```
In [19]: # You can find a similar image as per the images you have in the metadata
# In this case, you have a table (picked from the same document source) a
image_query_path = "tac_table_revenue.png"

# Print a message indicating the input image
print("***Input image from user:***")

# Display the input image
PIL_Image.open(image_query_path)
```

```
***Input image from user:***
```

```
Out[19]:
```

	Year Ended December 31,	
	2020	2021
TAC	\$ 32,778	\$ 45,566
Other cost of revenues	51,954	65,373
Total cost of revenues	\$ 84,732	\$ 110,939
Total cost of revenues as a percentage of revenues	46.4 %	43.1 %

You expect to find tables (as images) that are similar in terms of "Other/Total cost of revenues."

```
In [20]: # Search for Similar Images Based on Input Image and Image Embedding

matching_results_image = get_similar_image_from_query(
    text_metadata_df,
    image_metadata_df,
    query=query, # Use query text for additional filtering (optional)
    column_name="mm_embedding_from_img_only", # Use image embedding for
    image_emb=True,
    image_query_path=image_query_path, # Use input image for similarity
    top_n=3, # Retrieve top 3 matching images
    embedding_size=1408, # Use embedding size of 1408
)
```

```
print("\n **** Result: ***** \n")

# Display the Top Matching Image
display(
    matching_results_image[0]["image_object"]
) # Display the top matching image object (Pillow Image)
```

**** Result: ****

	Year Ended December 31,	
	2020	2021
TAC	\$ 32,778	\$ 45,566
Other cost of revenues	51,954	65,373
Total cost of revenues	\$ 84,732	\$ 110,939
Total cost of revenues as a percentage of revenues	46.4 %	43.1 %

It did find a similar-looking image (table), which gives more detail about different revenue, expenses, income, and a few more details based on the given image. More importantly, both tables show numbers related to the "cost of revenue."

You can also print the citation to see what it has matched.

```
In [21]: # Display citation details for the top matching image
print_text_to_image_citation(
    matching_results_image, print_top=True
) # Print citation details for the top matching image
```

Citation 1:

Matched image path, page number, and page text:

Score: 0.99

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_1_0_13.jpeg

Page number: 2

Page text: APAC revenue growth from 2020 to 2021 was favorably affected by foreign

currency exchange rates, primarily due to the U.S. dollar weakening relative to

the Australian dollar, partially offset by the U.S. dollar strengthening relative to

the Japanese yen.

Other Americas growth change from 2020 to 2021 was favorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar

weakening relative to the Canadian dollar, partially offset by the U.S. dollar

strengthening relative to the Argentine peso and the Brazilian real.

Costs and Expenses

Cost of Revenues

The following tables present cost of revenues, including TAC (in millions, except percentages):

Cost of revenues increased \$26.2 billion from 2020 to 2021. The increase was

due to an increase in other cost of revenues and TAC of \$13.4 billion and \$12.8

billion, respectively.

The increase in TAC from 2020 to 2021 was due to an increase in TAC paid to

distribution partners and to Google Network partners, primarily driven by growth

in revenues subject to TAC. The TAC rate decreased from 22.3% to 21.8% from

2020 to 2021 primarily due to a revenue mix shift from Google Network properties to Google Search & other properties.

The TAC rate on Google Search & other properties revenues and the TAC rate on Google Network revenues were both substantially consistent from 2020 to 2021. The increase in other cost of revenues from 2020 to 2021 was driven by

Image description: The image displays a financial table comparing the "Total cost of revenues" for the years ended December 31, 2020, and 2021. It breaks down these costs into "TAC" and "Other cost of revenues," and provides the total cost as a percentage of revenues for each year.

****Table Elements:****

* ****Year Ended December 31, 2020:****

* TAC: \$ 32,778

* Other cost of revenues: 51,954

* Total cost of revenues: \$ 84,732

* Total cost of revenues as a percentage of revenues: 46.4 %

* ****Year Ended December 31, 2021:****

* TAC: \$ 45,566

* Other cost of revenues: 65,373

* Total cost of revenues: \$ 110,939

* Total cost of revenues as a percentage of revenues: 43.1 %

```
In [22]: # Check Other Matched Images (Optional)
# You can access the other two matched images using:

print("-----Matched Images-----\n")
display_images(
    [
        matching_results_image[0]["img_path"],
        matching_results_image[1]["img_path"],
    ],
    resize_ratio=0.5,
)
```

-----Matched Images-----

	Year Ended December 31,	
	2020	2021
TAC	\$ 32,778	\$ 45,566
Other cost of revenues	51,954	65,373
Total cost of revenues	\$ 84,732	\$ 110,939
Total cost of revenues as a percentage of revenues	46.4 %	43.1 %

	Year Ended December 31,	
	2020	2021
United States	47 %	46 %
EMEA	30 %	31 %
APAC	18 %	18 %
Other Americas	5 %	5 %

The ability to identify similar text and images based on user input, using Gemini and embeddings, forms a crucial foundation for development of multimodal RAG systems, which you explore in the next section.

Comparative reasoning

Next, let's apply what you have done so far to doing comparative reasoning.

For this example:

Step 1: You will search all the images for a specific query

Step 2: Send those images to Gemini to ask multiple questions, where it has to compare and provide you with answers.

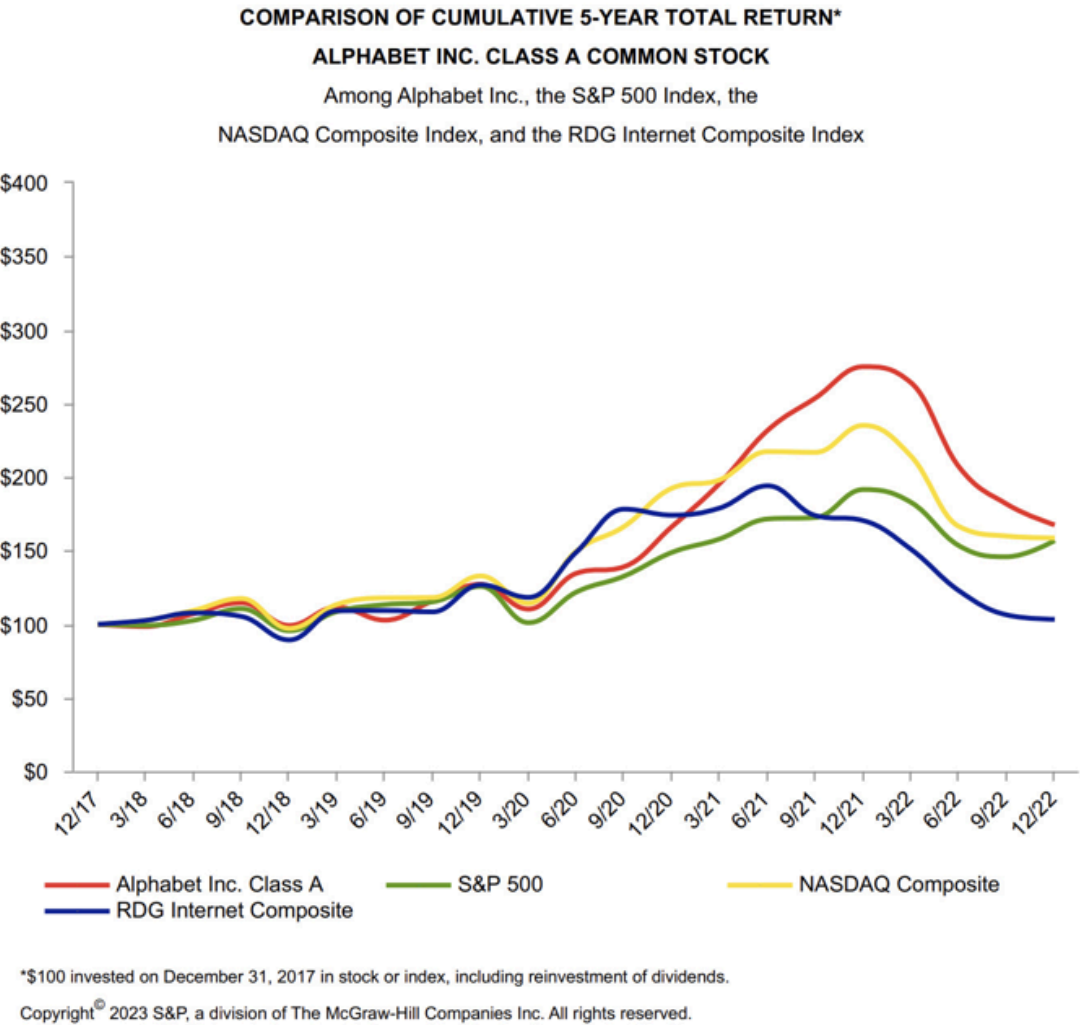
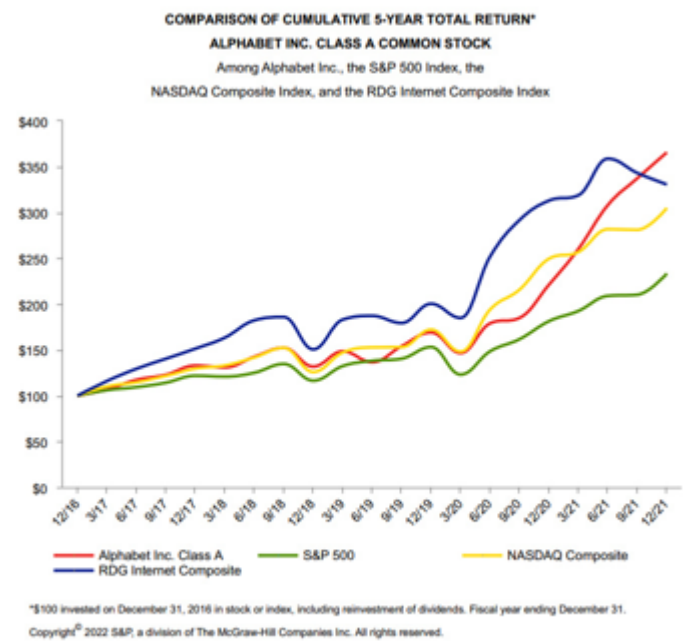
```
In [23]: matching_results_image_query_1 = get_similar_image_from_query(
    text_metadata_df,
    image_metadata_df,
    query="Show me all the graphs that shows Google Class A cumulative 5-
    column_name="text_embedding_from_image_description", # Use image des
    image_emb=False, # Use text embedding instead of image embedding
    top_n=3,
    embedding_size=1408,
)
```

```
In [24]: # Check Matched Images
# You can access the other two matched images using:

print("-----Matched Images-----\n")
display_images(
    [
        matching_results_image_query_1[0]["img_path"],
```

```
matching_results_image_query_1[1]["img_path"],
],
resize_ratio=0.5,
)
```

-----Matched Images-----



```
In [25]: prompt = f""" Instructions: Compare the images and the Gemini extracted t
Make sure to think thoroughly before answering the question and put the n

Context:
Image_1: {matching_results_image_query_1[0]["image_object"]}
gemini_extracted_text_1: {matching_results_image_query_1[0]['image_descri
Image_2: {matching_results_image_query_1[1]["image_object"]}
gemini_extracted_text_2: {matching_results_image_query_1[1]['image_descri

Question:
- Key findings of Class A share?
- What are the critical differences between the graphs for Class A Share
- What are the key findings of Class A shares concerning the S&P 500?
- Which index best matches Class A share performance closely where Googl
- Identify key chart patterns in both graphs.
- Which index best matches Class A share performance closely where Googl
"""

# Generate Gemini response with streaming output
rich_Markdown(
    get_gemini_response(
        model, # we are passing "gemini-3.1-flash-lite" model
        model_input=[prompt],
        stream=True,
        generation_config=types.GenerateContentConfig(temperature=1),
    )
)
```

Out[25]: To answer your questions based on the provided graphs and context,

1. Key Findings of Class

- **Graph 1 (2016–2021):** Alphabet Inc. Class A shares show a steady highest-performing investment of the four categories by the end
- **Graph 2 (2017–2022):** Alphabet Inc. shares experienced significant noticeable decline throughout 2022. Despite this decline, it retained value at the end of the period.

2. Critical Differences Between

- **Timeframes:** The graphs cover different five-year periods: Graph 1 covers Dec 31, 2016 – Dec 31, 2021; Graph 2 covers Dec 31, 2017 – Dec 31, 2022.
- **Performance Outcome:** Graph 1 reflects a period of consistent growth, while Graph 2 shows a market correction of 2022, showing a "peak and drop" pattern compared to Graph 1.

3. Key Findings: Class A Share

- In both graphs, Alphabet Inc. consistently outperformed the S&P 500.
- While the S&P 500 showed consistent growth, it remained the lowest performing index over the five-year period in Graph 1.
- In Graph 2, while the S&P 500 grew, by the end of 2022, it had not reached its starting value, whereas Alphabet maintained a higher overall return.

4. Index Closest to Class A Share Performance

- **The NASDAQ Composite (Yellow line):** This index tracks most closely Alphabet's performance.
- **Reasoning:** In Graph 2, the text explicitly notes that Alphabet's performance is closely aligned with the NASDAQ-100 index, starting in 2020. By the end of 2022, they ended at "very similar levels to one another," indicating a high correlation in their performance patterns over that period.

5. Key Chart Patterns

- **Growth/Bullish Trend:** Both graphs display an overall upward trend in share value.
- **Divergence:** In both graphs, a noticeable separation occurs after 2020 as Alphabet/NASDAQ begin to pull away from broader/older market indices.
- **Correction Pattern:** Graph 2 specifically demonstrates a "Mean Reversion" pattern where the sharp upward trajectory of 2020–2021 is followed by a downward correction in 2022.

6. Index Closest to Class A Share Performance

- **The NASDAQ Composite Index:** As established in step 4, the NASDAQ Composite is the closest index to Class A Share performance.
- **Reasoning:** Since Alphabet is a large-cap technology stock, its performance profile aligns most closely with the NASDAQ-100 (which is heavily tech-focused) rather than the S&P 500 or the specific RDG Internet Composite, which likely has different sector profiles.

⚠ **Disclaimer: This is not a real investment advice and should not be taken seriously!!** ⚠

Multimodal retrieval augmented generation (RAG)

Let's bring everything together to implement multimodal RAG. You will use all the elements that you've explored in previous sections to implement the multimodal RAG.

These are the steps:

- **Step 1:** The user gives a query in text format where the expected information is available in the document and is embedded in images and text.
- **Step 2:** Find all text chunks from the pages in the documents using a method similar to the one you explored in `Text Search`.
- **Step 3:** Find all similar images from the pages based on the user query matched with `image_description` using a method identical to the one you explored in `Image Search`.
- **Step 4:** Combine all similar text and images found in steps 2 and 3 as `context_text` and `context_images`.
- **Step 5:** With the help of Gemini, we can pass the user query with text and image context found in steps 2 & 3. You can also add a specific instruction the model should remember while answering the user query.
- **Step 6:** Gemini produces the answer, and you can print the citations to check all relevant text and images used to address the query.

Step 1: User query

```
In [26]: # this time we are not passing any images, but just a simple text query.

query = """Questions:
- What are the critical difference between various graphs for Class A Sh
- Which index best matches Class A share performance closely where Googl
- Identify key chart patterns for Google Class A shares.
- What is cost of revenues, operating expenses and net income for 2020.
- What was the effect of Covid in the 2020 financial year?
- What are the total revenues for APAC and USA for 2021?
- What is deferred income taxes?
- How do you compute net income per share?
- What drove percentage change in the consolidated revenue and cost of r
- What is the cause of 41% increase in revenue from 2020 to 2021 and how
"""
```

Step 2: Get all relevant text chunks

```
In [27]: # Retrieve relevant chunks of text based on the query
matching_results_chunks_data = get_similar_text_from_query(
    query,
    text_metadata_df,
    column_name="text_embedding_chunk",
    top_n=10,
    chunk_text=True,
)
```

Step 3: Get all relevant images

```
In [28]: # Get all relevant images based on user query
matching_results_image_fromdescription_data = get_similar_image_from_query(
    text_metadata_df,
    image_metadata_df,
    query=query,
    column_name="text_embedding_from_image_description",
    image_emb=False,
```

```

    top_n=10,
    embedding_size=1408,
)

```

Step 4: Create context_text and context_images

```

In [29]: # combine all the selected relevant text chunks
context_text = []
for key, value in matching_results_chunks_data.items():
    context_text.append(value["chunk_text"])
final_context_text = "\n".join(context_text)

# combine all the relevant images and their description generated by Gemini
context_images = []
for key, value in matching_results_image_fromdescription_data.items():
    context_images.extend(
        ["Image: ", value["image_object"], "Caption: ", value["image_desc
    )

```

Step 5: Pass context to Gemini

```

In [30]: prompt = f""" Instructions: Compare the images and the text provided as C
Make sure to think thoroughly before answering the question and put the n
If unsure, respond, "Not enough context to answer".

Context:
- Text Context:
{final_context_text}
- Image Context:
{context_images}

{query}

Answer:
"""

# Generate Gemini response with streaming output
rich_Markdown(
    get_gemini_response(
        model,
        model_input=[prompt],
        stream=True,
        generation_config=types.GenerateContentConfig(temperature=1),
    )
)

```

Out[30]: Here are the answers to your questions based on the provided conte

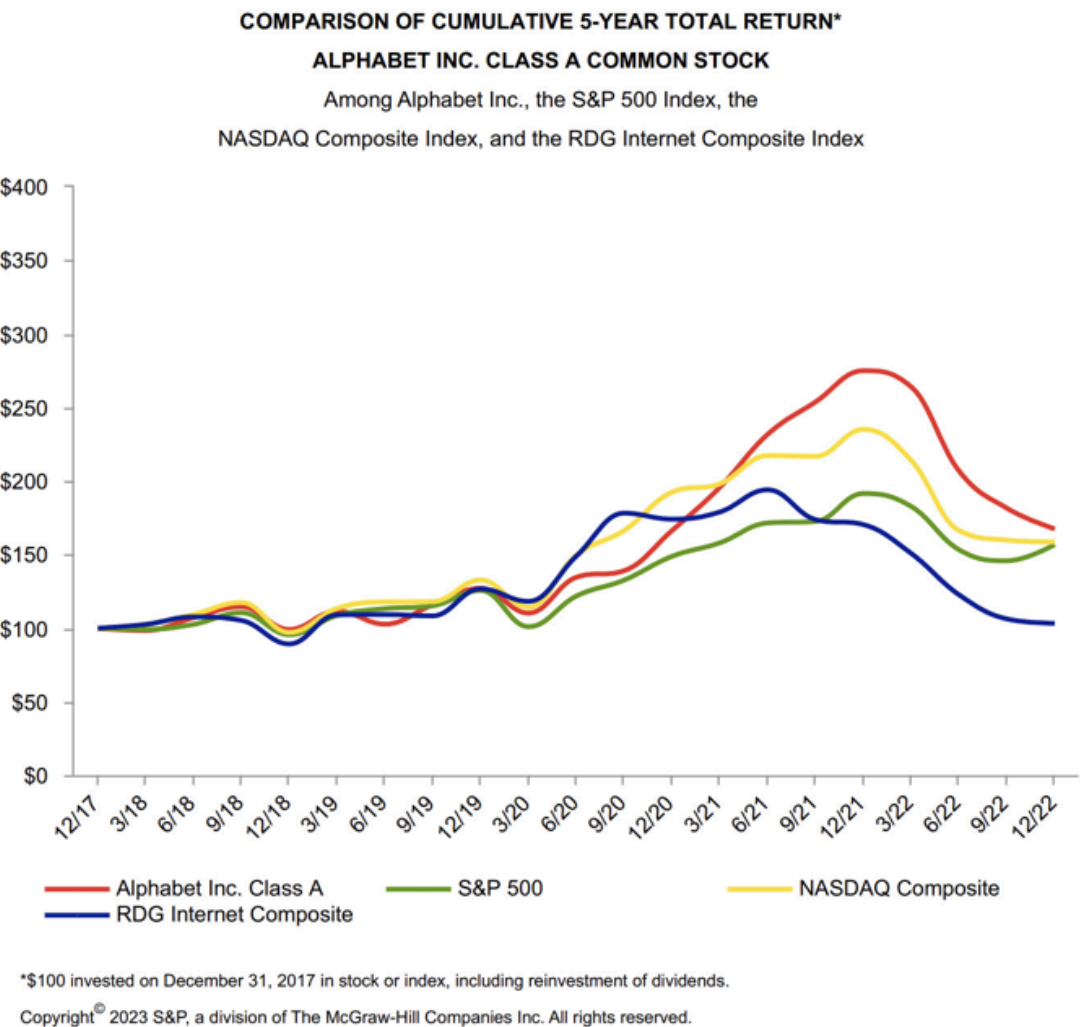
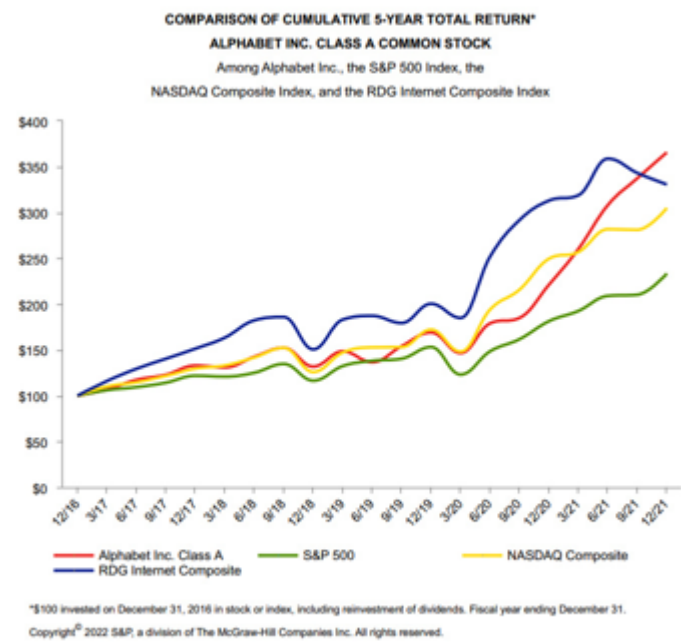
- **What are the critical difference between various graphs for Cla**
 - The first graph (ending Dec 31, 2021) covers a 5-year period the highest-performing investment, with the NASDAQ and RDG i lowest.
 - The second graph (ending Dec 31, 2022) covers a 5-year perio 2022 market decline. It shows that while Alphabet and the NA they finished the 5-year window at a similar level to each o Composite also trended toward each other at lower values.
- **Which index best matches Class A share performance closely wher reasoning.**
 - Not enough context to answer. The document does not provide 500, NASDAQ Composite, RDG Internet Composite) to determine
- **Identify key chart patterns for Google Class A shares.**
 - Both graphs show a consistent upward growth trend from 2017
 - A significant peak is observed around late 2021, followed by (as shown in the second graph).
- **What is cost of revenues, operating expenses and net income for**
 - Cost of revenues: \$84,732 million (31% increase in 2021).
 - Operating expenses: \$56,571 million (20% increase in 2021).
 - Net Income: \$40,269 million (89% increase in 2021).
- **What was the effect of Covid in the 2020 financial year?**
 - Advertising revenues declined starting in March 2020 due to non-commercial topics and reduced advertiser spending.
 - This decline continued through the quarter ended June 30, 20
 - Adversely affected 2020 advertising revenues, which subseque seen in 2021 as the economy recovered.
- **What are the total revenues for APAC and USA for 2021?**
 - APAC revenues: \$46,123 million.
 - United States revenues: \$117,854 million.
- **What is deferred income taxes?**
 - The context provides a table detailing "Deferred tax assets" accrued benefits) and "Deferred tax liabilities" (e.g., prop the net temporary differences between the carrying amount of their tax bases.
- **How do you compute net income per share?**
 - Alphabet uses the two-class method.
 - Basic EPS: Net income (undistributed earnings) divided by th
 - Diluted EPS: Uses the weighted-average number of shares plus securities (restricted stock units and other contingently is
- **What drove percentage change in the consolidated revenue and co effect of Covid?**
 - Revenue increase: Driven by Google Services, Google Cloud, a COVID-19 on 2020 advertising revenues.
 - Cost of revenue increase: Driven by higher Traffic Acquisiti including content acquisition costs (YouTube), data center/o depreciation expense due to changes in server useful life pa
- **What is the cause of 41% increase in revenue from 2020 to 2021**
 - Dollar change: \$75,110 million (\$257,637m in 2021 vs \$182,52
 - Causes: Growth in Google Services and Google Cloud, combined advertising decline in 2020.

Step 6: Print citations and references

```
In [31]: print("-----Matched Images-----\n")
display_images(
    [
        matching_results_image_fromdescription_data[0]["img_path"],
        matching_results_image_fromdescription_data[1]["img_path"],
        matching_results_image_fromdescription_data[2]["img_path"],
        matching_results_image_fromdescription_data[3]["img_path"],
```

```
],
resize_ratio=0.5,
)
```

-----Matched Images-----



	Year Ended December 31,		\$ Change	% Change
	2020	2021		
Consolidated revenues	\$ 182,527	\$ 257,637	\$ 75,110	41 %
Change in consolidated constant currency revenues				39 %
Cost of revenues	\$ 84,732	\$ 110,939	\$ 26,207	31 %
Operating expenses	\$ 56,571	\$ 67,984	\$ 11,413	20 %
Operating income	\$ 41,224	\$ 78,714	\$ 37,490	91 %
Operating margin	23 %	31 %		8 %
Other income (expense), net	\$ 6,858	\$ 12,020	\$ 5,162	75 %
Net Income	\$ 40,269	\$ 76,033	\$ 35,764	89 %
Diluted EPS	\$ 58.61	\$ 112.20	\$ 53.59	91 %
Number of Employees	135,301	156,500	21,199	16 %

	Year Ended December 31,		% Change from Prior Year
	2020	2021	
EMEA revenues	\$ 55,370	\$ 79,107	43 %
EMEA constant currency revenues		76,321	38 %
APAC revenues	32,550	46,123	42 %
APAC constant currency revenues		45,666	40 %
Other Americas revenues	9,417	14,404	53 %
Other Americas constant currency revenues		14,317	52 %
United States revenues	85,014	117,854	39 %
Hedging gains (losses)	176	149	
Total revenues	\$ 182,527	\$ 257,637	41 %
Revenues, excluding hedging effect	\$ 182,351	\$ 257,488	
Exchange rate effect		(3,330)	
Total constant currency revenues		\$ 254,158	39 %

```
In [32]: # Image citations. You can check how Gemini generated metadata helped in
print_text_to_image_citation(
    matching_results_image_fromdescription_data, print_top=False
)
```

Citation 1:

Matched image path, page number, and page text:

Score: 0.71

File name: google-10k-sample-part1.pdf

Path: images/google-10k-sample-part1.pdf_image_2_0_15.jpeg

Page number: 3

Page text: Stock Performance Graphs

The graph below matches Alphabet Inc. Class A's cumulative 5-year total stockholder return on common stock with the cumulative total returns of the S&P

500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a \$100 investment in our common stock and in each index (with the reinvestment of all dividends) from December

31, 2016 to December 31, 2021. The returns shown are based on historical results and are not intended to suggest future performance.

Image description: The image is a line graph titled "Comparison of Cumulative 5-Year Total Return" for Alphabet Inc. Class A Common Stock. It tracks the performance of an initial investment made on December 31, 2016, comparing Alphabet Inc. against the S&P 500 Index, the NASDAQ Composite Index, and the RDG Internet Composite Index over a five-year period ending December 31, 2021.

****Findings from the graph:****

* ****Alphabet Inc. Class A (Red line):**** Shows a steady upward trend throughout the period, finishing as the highest-performing investment among the four categories by the end of 2021.

* ****RDG Internet Composite Index (Blue line):**** Experienced the most significant growth for the majority of the period, peaking around mid-2021 before experiencing a slight decline toward the end of the year.

* ****NASDAQ Composite Index (Yellow line):**** Followed a similar growth trajectory to the other indices, maintaining a position between the RDG Internet Composite and the S&P 500 for most of the timeframe.

* ****S&P 500 Index (Green line):**** Remained the lowest-performing investment of the four throughout the entire five-year period, though it still showed consistent growth from the initial investment point.

All four investments began at the same starting point on December 31, 2016, and all showed positive cumulative returns by the end of the five-year period.

Citation 2:

Matched image path, page number, and page text:

Score: 0.69

File name: google-10k-sample-part1.pdf

Path: images/google-10k-sample-part1.pdf_image_3_0_18.jpeg

Page number: 4

Page text: The graph below matches Alphabet Inc. Class A's cumulative 5-year total

stockholder return on common stock with the cumulative total returns of the S&P

500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a \$100 investment in our common stock and in each index (with the reinvestment of all dividends) from December

31, 2017 to December 31, 2022. The returns shown are based on historical results and are not intended to suggest future performance.

Image description: The image is a line graph titled "COMPARISON OF CUMULAT

IVE 5-YEAR TOTAL RETURN* ALPHABET INC. CLASS A COMMON STOCK." It compares the performance of Alphabet Inc. Class A Common Stock against the S&P 500 Index, the NASDAQ Composite Index, and the RDG Internet Composite Index over a five-year period starting from December 31, 2017.

****Findings from the graph:****

* ****Starting Point:**** All four investments began at a value of \$100 on December 31, 2017.

* ****Performance Trends:****

* From late 2017 through 2019, all four indices and the stock tracked relatively closely to one another.

* Starting in 2020, Alphabet Inc. (red line) and the NASDAQ Composite (yellow line) began to outperform the S&P 500 (green line) and the RDG Internet Composite (blue line).

* Alphabet Inc. reached the highest peak of the four lines around the end of 2021.

* Following the peak in late 2021, all four lines experienced a downward trend throughout 2022.

* ****Ending Point (December 2022):**** By the end of the period, Alphabet Inc. (red line) and the NASDAQ Composite (yellow line) remained higher than their initial starting value, while the S&P 500 (green line) and the RDG Internet Composite (blue line) ended closer to their original starting value. Alphabet Inc. and the NASDAQ Composite ended at very similar levels to one another, while the S&P 500 and the RDG Internet Composite also ended at similar levels to one another.

Citation 3:

Matched image path, page number, and page text:

Score: 0.68

File name: google-10k-sample-part1.pdf

Path: images/google-10k-sample-part1.pdf_image_4_0_21.jpeg

Page number: 5

Page text: Executive Overview

The following table summarizes consolidated financial results for the years ended

December 31, 2020 and 2021 unless otherwise specified (in millions, except for per share information and percentages):

Revenues were \$257.6 billion, an increase of 41%. The increase in revenues was primarily driven by Google Services and Google Cloud. The adverse effect of COVID-19 on 2020 advertising revenues also contributed to the year-over-year growth.

Cost of revenues was \$110.9 billion, an increase of 31%, primarily driven by increases in TAC and content acquisition costs.

An overall increase in data centers and other operations costs was partially

offset by a reduction in depreciation expense due to the change in the estimated useful life of our servers and certain network equipment.

Operating expenses were \$68.0 billion, an increase of 20%, primarily driven by headcount growth, increases in advertising and promotional expenses and charges related to legal matters.

Image description: The image displays a financial performance table comparing various metrics for the years ended December 31, 2020, and 2021, including the dollar and percentage changes between those two years.

Table Data

Metric	2020	2021	\$ Change	% Change
	---	---	---	---

Consolidated revenues	\$ 182,527	\$ 257,637	\$ 75,110	41 %
Change in consolidated constant currency revenues				39 %
Cost of revenues	\$ 84,732	\$ 110,939	\$ 26,207	31 %
Operating expenses	\$ 56,571	\$ 67,984	\$ 11,413	20 %
Operating income	\$ 41,224	\$ 78,714	\$ 37,490	91 %
Operating margin	23 %	31 %	8 %	
Other income (expense), net	\$ 6,858	\$ 12,020	\$ 5,162	75 %
Net Income	\$ 40,269	\$ 76,033	\$ 35,764	89 %
Diluted EPS	\$ 58.61	\$ 112.20	\$ 53.59	91 %
Number of Employees	135,301	156,500	21,199	16 %

Citation 4:

Matched image path, page number, and page text:

Score: 0.68

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_0_1_8.jpeg

Page number: 1

Page text: source: https://abc.xyz/assets/investor/static/pdf/20220202_alp_habet_10K.pdf

source: <https://abc.xyz/assets/9a/bd/838c917c4b4ab21f94e84c3c2c65/goog-10-k-q4-2022.pdf>

Note: Tables and figures are converted to images for demonstration purpose s.

Revenues by Geography

The following table presents revenues by geography as a percentage of revenues, determined based on the addresses of our customers:

The following table presents the foreign exchange effect on international revenues and total revenues (in millions, except percentages):

EMEA revenue growth from 2020 to 2021 was favorably affected by foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Euro and British pound.

Image description: The image displays a financial table comparing revenues across different geographic regions and categories for the years ended December 31, 2020, and 2021, along with the percentage change from the prior year.

Table Data

Category	2020	2021	% Change from Prior Year
EMEA revenues	55,370	79,107	43%
EMEA constant currency revenues		76,321	38%
APAC revenues	32,550	46,123	42%
APAC constant currency revenues		45,666	40%
Other Americas revenues	9,417	14,404	53%
Other Americas constant currency revenues		14,317	52%
United States revenues	85,014	117,854	39%
Hedging gains (losses)	176	149	
Total revenues	**182,527**	**257,637**	**41%**
Revenues, excluding hedging effect	182,351	257,488	
Exchange rate effect	(3,330)		
Total constant currency revenues	**254,158**	**39%**	

Citation 5:

Matched image path, page number, and page text:

Score: 0.66

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_4_0_22.jpeg

Page number: 5

Page text: Stock-Based Award Activities

The weighted-average grant-date fair value of RSUs granted during the year ended December 31, 2019 and 2020 was \$1,092.36 and \$1,407.97, respectively. Total fair value of RSUs, as of their respective vesting dates, during the years ended December 31, 2019, 2020, and 2021 were \$15.2 billion, \$17.8 billion, and \$28.8 billion, respectively. As of December 31, 2021, there was \$25.8 billion of unrecognized compensation cost related to unvested employee RSUs. This amount is expected to be recognized over a weighted-average period of 2.5 years. 401(k) Plans We have two 401(k) Savings Plans that qualify as deferred

Image description: The image displays a financial table detailing the calculation of basic and diluted net income per share for three different classes of stock (Class A, Class B, and Class C) for the year ended December 31, 2020.

Table Data

Basic net income per share:

* ****Numerator:****
 * Allocation of undistributed earnings: Class A (\$ 17,733), Class B (\$ 2,732), Class C (\$ 19,804)
 * ****Denominator:****
 * Number of shares used in per share computation: Class A (299,815), Class B (46,182), Class C (334,819)
 * ****Basic net income per share:**** Class A (\$ 59.15), Class B (\$ 59.15), Class C (\$ 59.15)

Diluted net income per share:

* ****Numerator:****
 * Allocation of undistributed earnings for basic computation: Class A (\$ 17,733), Class B (\$ 2,732), Class C (\$ 19,804)
 * Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares: Class A (2,732), Class B (0), Class C (0)
 * Reallocation of undistributed earnings: Class A ((180)), Class B ((25)), Class C (180)
 * Allocation of undistributed earnings: Class A (\$ 20,285), Class B (\$ 2,707), Class C (\$ 19,984)
 * ****Denominator:****
 * Number of shares used in basic computation: Class A (299,815), Class B (46,182), Class C (334,819)
 * ****Weighted-average effect of dilutive securities:****
 * Add: Conversion of Class B to Class A shares outstanding: Class A (46,182), Class B (0), Class C (0)
 * Add: Restricted stock units and other contingently issuable shares: Class A (87), Class B (0), Class C (6,125)
 * Number of shares used in per share computation: Class A (346,084), Class B (46,182), Class C (340,944)
 * ****Diluted net income per share:**** Class A (\$ 58.61), Class B (\$ 58.61), Class C (\$ 58.61)

Citation 6:

Matched image path, page number, and page text:

Score: 0.63

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_1_0_13.jpeg

Page number: 2

Page text: APAC revenue growth from 2020 to 2021 was favorably affected by foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Australian dollar, partially offset by the U.S. dollar strengthening relative to the Japanese yen.

Other Americas growth change from 2020 to 2021 was favorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Canadian dollar, partially offset by the U.S. dollar strengthening relative to the Argentine peso and the Brazilian real.

Costs and Expenses

Cost of Revenues

The following tables present cost of revenues, including TAC (in millions, except percentages):

Cost of revenues increased \$26.2 billion from 2020 to 2021. The increase was due to an increase in other cost of revenues and TAC of \$13.4 billion and \$12.8

billion, respectively.

The increase in TAC from 2020 to 2021 was due to an increase in TAC paid to distribution partners and to Google Network partners, primarily driven by growth

in revenues subject to TAC. The TAC rate decreased from 22.3% to 21.8% from

2020 to 2021 primarily due to a revenue mix shift from Google Network properties to Google Search & other properties.

The TAC rate on Google Search & other properties revenues and the TAC rate on Google Network revenues were both substantially consistent from 2020 to 2021. The increase in other cost of revenues from 2020 to 2021 was driven by

Image description: The image displays a financial table comparing the "Total cost of revenues" for the years ended December 31, 2020, and 2021. It breaks down these costs into "TAC" and "Other cost of revenues," and provides the total cost as a percentage of revenues for each year.

****Table Elements:****

* ****Year Ended December 31, 2020:****

* TAC: \$ 32,778

* Other cost of revenues: 51,954

* Total cost of revenues: \$ 84,732

* Total cost of revenues as a percentage of revenues: 46.4 %

* ****Year Ended December 31, 2021:****

* TAC: \$ 45,566

* Other cost of revenues: 65,373

* Total cost of revenues: \$ 110,939

* Total cost of revenues as a percentage of revenues: 43.1 %

Citation 7:

Matched image path, page number, and page text:

Score: 0.63

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_3_0_19.jpeg

Page number: 4

Page text: liquidation and dividend rights are identical, the undistributed earnings are allocated on a proportionate basis. In the years ended December 31, 2019, 2020 and 2021, the net income per share amounts are the same for Class A, Class B, and Class C stock because the holders of each class are entitled to equal per share dividends or distributions in liquidation in accordance with the Amended and Restated Certificate of Incorporation of Alphabet Inc. The following tables set forth the computation of basic and diluted net income per share of Class A, Class B, and Class C stock (in millions, except share amounts which are reflected in thousands and per share amounts):

Image description: The image displays a financial table detailing the calculation of basic and diluted net income per share for three different classes of stock (Class A, Class B, and Class C) for the year ended December 31, 2019.

Table Data

****Basic net income per share:****

* ****Numerator:****

* Allocation of undistributed earnings: Class A (\$ 14,846), Class B (\$ 2,307), Class C (\$ 17,190)

* ****Denominator:****

* Number of shares used in per share computation: Class A (299,402), Class B (46,527), Class C (346,667)

* ****Basic net income per share:**** Class A (\$ 49.59), Class B (\$ 49.59), Class C (\$ 49.59)

****Diluted net income per share:****

* ****Numerator:****

* Allocation of undistributed earnings for basic computation: Class A (\$ 14,846), Class B (\$ 2,307), Class C (\$ 17,190)

* Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares: Class A (2,307), Class B (0), Class C (0)

* Reallocation of undistributed earnings: Class A ((126)), Class B ((20)), Class C (126)

* Allocation of undistributed earnings: Class A (\$ 17,027), Class B (\$ 2,287), Class C (\$ 17,316)

* ****Denominator:****

* Number of shares used in basic computation: Class A (299,402), Class B (46,527), Class C (346,667)

* ****Weighted-average effect of dilutive securities:****

* Add: Conversion of Class B to Class A shares outstanding: Class A (46,527), Class B (0), Class C (0)

* Add: Restricted stock units and other contingently issuable shares: Class A (413), Class B (0), Class C (5,547)

* Number of shares used in per share computation: Class A (346,342), Class B (46,527), Class C (352,214)

* ****Diluted net income per share:**** Class A (\$ 49.16), Class B (\$ 49.16), Class C (\$ 49.16)

Citation 8:

Matched image path, page number, and page text:

Score: 0.62

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_0_0_6.jpeg

Page number: 1

Page text: source: https://abc.xyz/assets/investor/static/pdf/20220202_alphabet_10K.pdf

source: <https://abc.xyz/assets/9a/bd/838c917c4b4ab21f94e84c3c2c65/goog-10k-q4-2022.pdf>

Note: Tables and figures are converted to images for demonstration purposes.

Revenues by Geography

The following table presents revenues by geography as a percentage of revenues, determined based on the addresses of our customers:

The following table presents the foreign exchange effect on international revenues and total revenues (in millions, except percentages):

EMEA revenue growth from 2020 to 2021 was favorably affected by foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Euro and British pound.

Image description: The image displays a table showing the percentage distribution of a metric across different geographic regions for the years ended December 31, 2020, and 2021.

****Table Elements:****

Region	2020	2021
United States	47 %	46 %
EMEA	30 %	31 %
APAC	18 %	18 %
Other Americas	5 %	5 %

Citation 9:

Matched image path, page number, and page text:

Score: 0.61

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_4_1_23.jpeg

Page number: 5

Page text: Stock-Based Award Activities

The weighted-average grant-date fair value of RSUs granted during the year ended December 31, 2019 and 2020 was \$1,092.36 and \$1,407.97, respectively.

Total fair value of RSUs, as of their respective vesting dates, during the years

ended December 31, 2019, 2020, and 2021 were \$15.2 billion, \$17.8 billion, and

\$28.8 billion, respectively. As of December 31, 2021, there was \$25.8 billion of

unrecognized compensation cost related to unvested employee RSUs. This amount is expected to be recognized over a weighted-average period of 2.5 years. 401(k) Plans We have two 401(k) Savings Plans that qualify as deferred

Image description: The image displays a table summarizing the activities for or unvested Alphabet Restricted Stock Units (RSUs) for the year ended December 31, 2021.

Table Data

Activity	Number of Shares	Weighted-Average Grant-Date Fair Value
Unvested as of December 31, 2020	19,288,793	\$ 1,262.13
Granted	10,582,700	\$ 1,949.16

Vested	(11,209,486)	\$ 1,345.98
Forfeited/canceled	(1,767,294)	\$ 1,425.48
Unvested as of December 31, 2021	16,894,713	\$ 1,626.13

Citation 10:

Matched image path, page number, and page text:

Score: 0.6

File name: google-10k-sample-part2.pdf

Path: images/google-10k-sample-part2.pdf_image_5_0_26.jpeg

Page number: 6

Page text: salary arrangements under Section 401(k) of the Internal Revenue Code. Under

these 401(k) Plans, matching contributions are based upon the amount of the

employees contributions subject to certain limitations. We recognized expense of

approximately \$724 million, \$855 million, and \$916 million for the years ended

December 31, 2019, 2020, and 2021, respectively. Note 14. Income Taxes

Income from continuing operations before income taxes consisted of the following (in millions):

Deferred Income Taxes

Deferred income taxes reflect the net effects of temporary differences between

the carrying amounts of assets and liabilities for financial reporting purposes and

the amounts used for income tax purposes. Significant components of our deferred tax assets and liabilities were as follows (in millions):

As of December 31, 2021, our federal, state, and foreign net operating losses

carryforwards for income tax purposes were approximately \$5.6 billion, \$4.6

billion, and \$1.7 billion respectively. If not utilized, the federal net operating loss

carryforwards will begin to expire in 2023, foreign net operating loss

carryforwards will begin to expire in 2025 and the state net operating losses

carryforwards will begin to expire in 2028. It is more likely than not that certain

Image description: The image displays a financial table detailing the components of deferred tax assets and deferred tax liabilities as of December 31, 2020, and December 31, 2021.

Table Data

Category	2020	2021
Deferred tax assets:		
Accrued employee benefits	580	549
Accruals and reserves not currently deductible	1,049	1,816
Tax credits	3,723	5,179
Net operating losses	1,085	1,790
Operating leases	2,620	2,503
Intangible assets	1,525	2,034
Other	981	925
Total deferred tax assets	11,563	14,796
Valuation allowance	(4,823)	(7,129)
Total deferred tax assets net of valuation allowance	6,740	7,667
Deferred tax liabilities:		

```
| Property and equipment, net | (3,382) | (5,237) |  
| Net investment gains | (1,901) | (3,229) |  
| Operating leases | (2,354) | (2,228) |  
| Other | (1,580) | (946) |  
| **Total deferred tax liabilities** | **(9,217)** | **(11,640)** |  
| **Net deferred tax assets (liabilities)** | **(2,477)** | **(3,973)** |
```

In [33]: *# Text citations*

```
print_text_to_text_citation(  
    matching_results_chunks_data,  
    print_top=False,  
    chunk_text=True,  
)
```

Citation 1: Matched text:**Score:** 0.79**File name:** google-10k-sample-part1.pdf**Page number:** 5**Chunk number:** 1**Chunk text:** Executive Overview

The following table summarizes consolidated financial results for the years ended

December 31, 2020 and 2021 unless otherwise specified (in millions, except for

per share information and percentages):

Revenues were \$257.6 billion, an increase of 41%. The increase in revenues was primarily driven by Google Services and Google Cloud. The adverse effect of COVID-19 on 2020 advertising revenues also contributed to the year-over-year growth.

Cost of revenues was \$110.9 billion, an increase of 31%, primarily driven by increases in TAC and content acquisition costs.

An overall increase in data centers and other operations costs was partially

offset by a reduction in depreciation expense due to the change in the estimated useful life of our servers and certain network equipment.

Operating expenses were \$68.0 billion, an increase of 20%, primarily driven by headcount growth, increases in advertising and promotional expenses and charges related to legal matters.

Citation 2: Matched text:**Score:** 0.76**File name:** google-10k-sample-part2.pdf**Page number:** 3**Chunk number:** 1

Chunk text: increases in content acquisition costs primarily for YouTube, data center and other operations costs, and hardware costs. The increase in data center and

Table of Contents Alphabet Inc. 36 other operations costs was partially offset by

a reduction in depreciation expense due to the change in the estimated useful life

of our servers and certain network equipment beginning in the first quarter of

2021.

Net Income Per Share

We compute net income per share of Class A, Class B, and Class C stock using

the two-class method. Basic net income per share is computed using the weighted-average number of shares outstanding during the period. Diluted net

income per share is computed using the weighted-average number of shares and

the effect of potentially dilutive securities outstanding during the period.

Potentially dilutive securities consist of restricted stock units and other

contingently issuable shares. The dilutive effect of outstanding restricted stock

units and other contingently issuable

Citation 3: Matched text:**Score:** 0.75**File name:** google-10k-sample-part2.pdf**Page number:** 2

Chunk number: 1

Chunk text: APAC revenue growth from 2020 to 2021 was favorably affected by foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Australian dollar, partially offset by the U.S. dollar strengthening relative to the Japanese yen.

Other Americas growth change from 2020 to 2021 was favorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Canadian dollar, partially offset by the U.S. dollar strengthening relative to the Argentine peso and the Brazilian real.

Costs and Expenses

Cost of Revenues

The following tables present cost of revenues, including TAC (in millions, except percentages):

Cost of revenues increased \$26.2 billion from 2020 to 2021. The increase was due to an increase in other cost of revenues and TAC of \$13.4 billion and \$12.8 billion, respectively.

The increase in TAC from 2020 to 2021 was due to an increase in TAC paid to distribution partners and to Google Network partners, primarily

Citation 4: Matched text:

Score: 0.73

File name: google-10k-sample-part2.pdf

Page number: 2

Chunk number: 2

Chunk text: 21 was due to an increase in TAC paid to distribution partners and to Google Network partners, primarily driven by growth in revenues subject to TAC. The TAC rate decreased from 22.3% to 21.8% from

2020 to 2021 primarily due to a revenue mix shift from Google Network properties to Google Search & other properties.

The TAC rate on Google Search & other properties revenues and the TAC rate on Google Network revenues were both substantially consistent from 2020 to 2021. The increase in other cost of revenues from 2020 to 2021 was driven by

Citation 5: Matched text:

Score: 0.73

File name: google-10k-sample-part1.pdf

Page number: 3

Chunk number: 1

Chunk text: Stock Performance Graphs

The graph below matches Alphabet Inc. Class A's cumulative 5-year total stockholder return on common stock with the cumulative total returns of the S&P

500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a \$100 investment in our common stock and in each index (with the reinvestment of all dividends) from December

31, 2016 to December 31, 2021. The returns shown are based on historical results and are not intended to suggest future performance.

Citation 6: Matched text:**Score:** 0.72**File name:** google-10k-sample-part1.pdf**Page number:** 6**Chunk number:** 2**Chunk text:** te 8 of the Notes to

Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K for further information.

On February 1, 2022, the Company announced that the Board of Directors had approved and declared a 20- for-one stock split in the form of a one-time special stock dividend on each share of the Companys Class A, Class B, and Class C stock. See Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K for additional information.

The Effect of COVID-19 on our Financial Results

We began to observe the effect of COVID-19 on our financial results in March

2020 when, despite an increase in users' search activity, our advertising revenues declined compared to the prior year. This was due to a shift of user

search activity to less commercial topics and reduced spending by our advertisers. For the quarter ended June 30, 2020 our advertising revenues declined due to the continued effects of COVID-19 and the rel

Citation 7: Matched text:**Score:** 0.72**File name:** google-10k-sample-part2.pdf**Page number:** 4**Chunk number:** 1

Chunk text: liquidation and dividend rights are identical, the undistributed earnings are allocated on a proportionate basis.

In the years ended December 31, 2019, 2020 and 2021, the net income per share amounts are the same for Class A, Class B, and Class C stock because the holders of each class are entitled to equal per share dividends or distributions

in liquidation in accordance with the Amended and Restated Certificate of Incorporation of Alphabet Inc.

The following tables set forth the computation of basic and diluted net income per

share of Class A, Class B, and Class C stock (in millions, except share amounts

which are reflected in thousands and per share amounts):

Citation 8: Matched text:**Score:** 0.71**File name:** google-10k-sample-part1.pdf**Page number:** 4**Chunk number:** 1

Chunk text: The graph below matches Alphabet Inc. Class A's cumulative 5-year total

stockholder return on common stock with the cumulative total returns of the S&P

500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a \$100 investment in our common stock and in each index (with the reinvestment of all dividends) from December

31, 2017 to December 31, 2022. The returns shown are based on historical results and are not intended to suggest future performance.

Citation 9: Matched text:**Score:** 0.7**File name:** google-10k-sample-part2.pdf**Page number:** 1**Chunk number:** 1**Chunk text:** source: https://abc.xyz/assets/investor/static/pdf/20220202_alphabet_10K.pdf

source: https://abc.xyz/assets/9a/bd/838c917c4b4ab21f94e84c3c2c65/goog-10-k-q4-2022.pdf

Note: Tables and figures are converted to images for demonstration purpose s.

Revenues by Geography

The following table presents revenues by geography as a percentage of revenues, determined based on the addresses of our customers:

The following table presents the foreign exchange effect on international revenues and total revenues (in millions, except percentages):

EMEA revenue growth from 2020 to 2021 was favorably affected by foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to

the Euro and British pound.

Citation 10: Matched text:**Score:** 0.7**File name:** google-10k-sample-part1.pdf**Page number:** 6**Chunk number:** 1**Chunk text:** Other information:

Operating cash flow was \$91.7 billion, primarily driven by revenues generated from our advertising products.

Share repurchases were \$50.3 billion, an increase of 62%. See Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K for further information.

Capital expenditures, which primarily reflected investments in technical infrastructure, were \$24.6 billion.

In January 2021, we updated the useful lives of certain of our servers and network equipment, resulting in a reduction in depreciation expense of \$2.6 billion recorded primarily in cost of revenues and R&D. See Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K for further information.

Our acquisition of Fitbit closed in early January 2021, and the related revenues are included in Google other. See Note 8 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on F

Conclusions

Congratulations on making it through this multimodal RAG notebook!

While multimodal RAG can be quite powerful, note that it can face some limitations:

- **Data dependency:** Needs high-quality paired text and visuals.
- **Computationally demanding:** Processing multimodal data is resource-intensive.
- **Domain specific:** Models trained on general data may not shine in specialized fields like medicine.

- **Black box:** Understanding how these models work can be tricky, hindering trust and adoption.

Despite these challenges, multimodal RAG represents a significant step towards search and retrieval systems that can handle diverse, multimodal data.